

Cargo-to-Door

Autonomous Last-Meter Delivery for Vertical Cities

A building-integrated logistics platform for new high-rises and historic retrofits.

[placeholder] Presented by: Team Name — PI, Co-PI, Members — Institution — Date

Agenda

1. Background

- Motivation & Problem Definition
- Literature Survey: state of the art, our proposed solution, breakthrough technologies

2. The Team & Innovative Product Development

- Team & capabilities, requirements, architecture, design, implementation, testing

3. Roadmap / Project Plan

- Milestones, deliverables, work packages, timeline

4. Commercialization & Business Model

- Market, competition, budget, risks

Part 1 — Background

Motivation & Problem Definition

- Last-mile is the broken link — ~30–50% of total e-commerce logistics cost, plus urban congestion and emissions.
- The "last-meter" problem: even at the lobby, the trip to the apartment door is manual, slow, and unreliable.
- Failed deliveries & porch piracy drive returns, theft, and customer churn.
- Vertical density breaks current solutions: drones cannot land on balconies, rovers cannot ride elevators, couriers cannot scale to 50-storey towers.
- Historic & protected buildings cannot be torn open for new infrastructure — yet residents demand the same service.

Goal: a continuous, autonomous, weatherproof path from sky (or sidewalk) to the resident's door — for new construction AND historic retrofits.

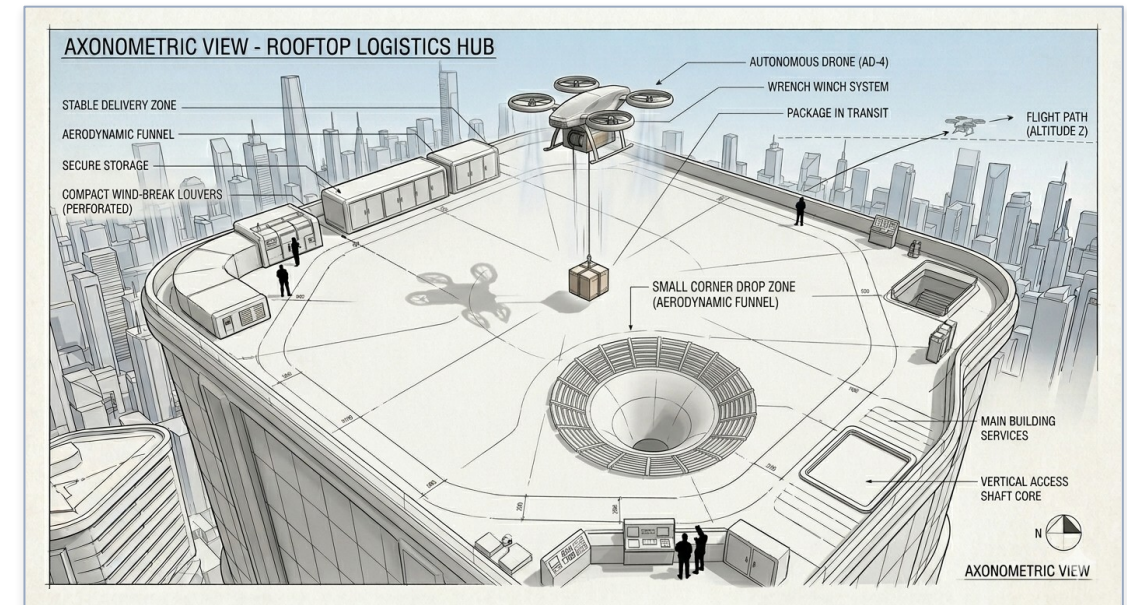
Literature Survey — Past Solutions / State of the Art

Approach	Examples	Strengths	Limitations
Drone-to-doorstep	Wing, Zipline, Matternet	Bypasses ground congestion	Needs landing pad; weather-sensitive; one parcel per flight; not vertical-city friendly
Sidewalk micro-rovers	Starship, Serve, Coco	Low-cost, social-friendly	Stuck at the lobby; no vertical access; theft risk
Parcel-locker arrays	Amazon Hub, InPost	Secure, asynchronous pickup	Still requires resident trip; ground-floor only
Pneumatic / vacuum tubes	Hospital legacy; Hyperloop-cargo concepts	Weather-immune, high throughput	Single-routing; expensive retrofit; small payload
In-building robotics	Relay, Savioke	Door-to-door inside building	Needs elevator integration; no exterior link

- Gap: no end-to-end system spans airspace -> building skin -> vertical core -> resident door, and none retrofits gracefully onto protected architecture.

Proposed Solution & Innovation

- Two parallel tracks, one platform.
- New builds — Rooftop Flight Deck + internal vertical shaft. Drones hover and winch-drop into an aerodynamic catchment funnel — no landing required.
- Historic builds — Exo-Logistics Spine: modular exterior pneumatic-decelerator tube clipped to the facade, feeding automatic floor-switchers inside.
- Passive-logistics philosophy: gravity chutes, motorized rollers, mechanical diverters and claws replace expensive robotic arms wherever possible.
- Multi-modal intake: same vertical core accepts both aerial drone and sidewalk micro-rover deliveries.



Breakthrough Technologies Required

- Hover-and-winch package drop — drone holds altitude Z while lowering a parcel; controller compensates for wind and tether sway.
- Aerodynamic catchment funnel + S-Trap drainage — captures off-center drops, lets rain through, blocks rain from following the parcel.
- Pneumatic decelerator tube — variable-pressure exterior shaft that slows a free-falling parcel from terminal velocity to safe handoff speed.
- Automatic floor-switcher — passive-actuated diverter that routes packages into the correct resident's intake chamber.
- Micro-rover <-> kinetic docking hatch — standardized mechanical & data interface so any six-wheeled rover can offload into the building.
- Building-level airspace orchestration — scheduling, geofencing, conflict-resolution for multiple drones over a single rooftop.

Part 2 — Team & Innovative Product

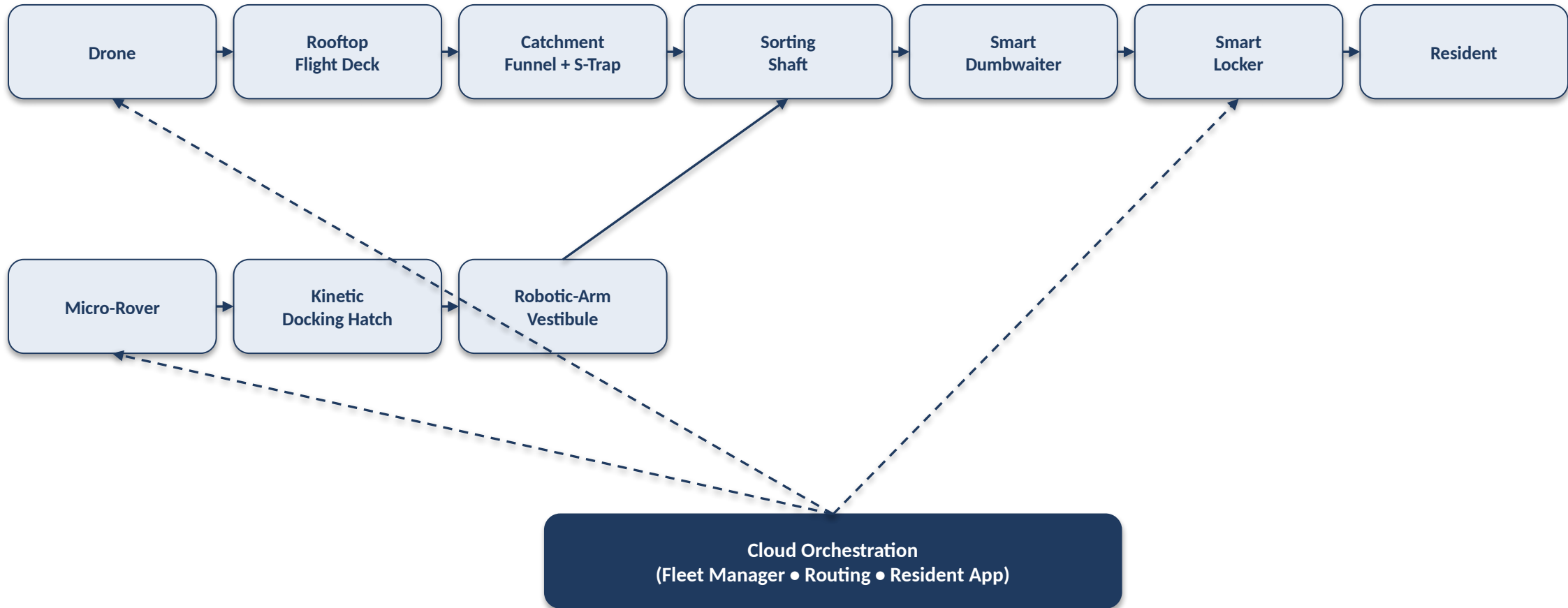
The Team & Capabilities

- *Fill with real names, affiliations, and CVs before submission.*

Role	Person	Capability brought
Principal Investigator	[name]	Systems engineering, autonomous vehicles
Mechanical Lead	[name]	Aerodynamic structures, pneumatic systems
Controls & Robotics	[name]	Drone control, robotic manipulation
Software & Cloud	[name]	Fleet orchestration, edge computing
Architecture & Civil	[name]	High-rise integration, historic-preservation retrofits
Business & Regulatory	[name]	Go-to-market, FAA / municipal liaison

- *Institutional capabilities: wind-tunnel access, drop-test rig, full-scale mockup hall, partnership LOIs with [developer] and [historic-district BID].*

Innovative Product / Process — System Overview



Two intake modalities (sky + street) converge on a single internal sorting core distributing to resident smart-lockers — orchestrated by the cloud layer.

Requirements Definition & Analysis

Functional

- *Throughput: ≥ 60 pkg/hr per building at peak*
- *Payload envelope: ≤ 5 kg, $\leq 40 \times 30 \times 30$ cm*
- *End-to-end latency drone-arrival -> locker: < 4 min*

Non-functional

- *Weatherproof to IP65 equivalent; operable in wind ≤ 12 m/s, rain ≤ 25 mm/h*
- *Acoustic limit at facade: ≤ 55 dBA at 10 m*
- *Fire-rating of internal shafts: ≥ 2 -hour*

Regulatory / Safety

- FAA Part 107 (or local equivalent) drone operations
- Local building, fire, and (for retrofits) landmark-commission codes

UX

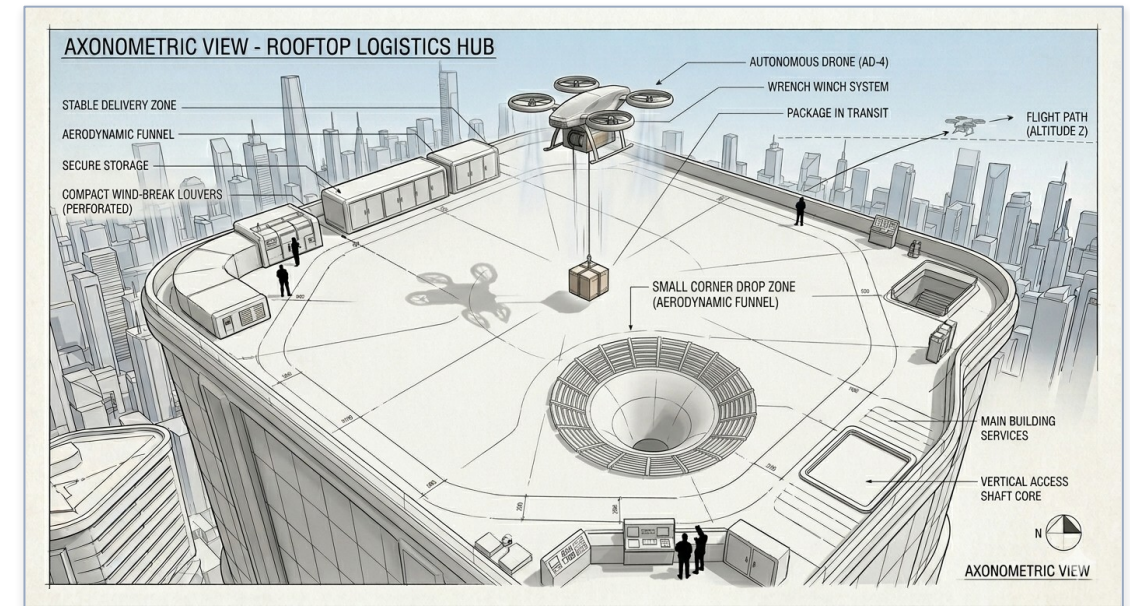
- *Resident retrieval at locker: < 30 s; mobile app notify within 10 s of locker close*

High-Level System Architecture



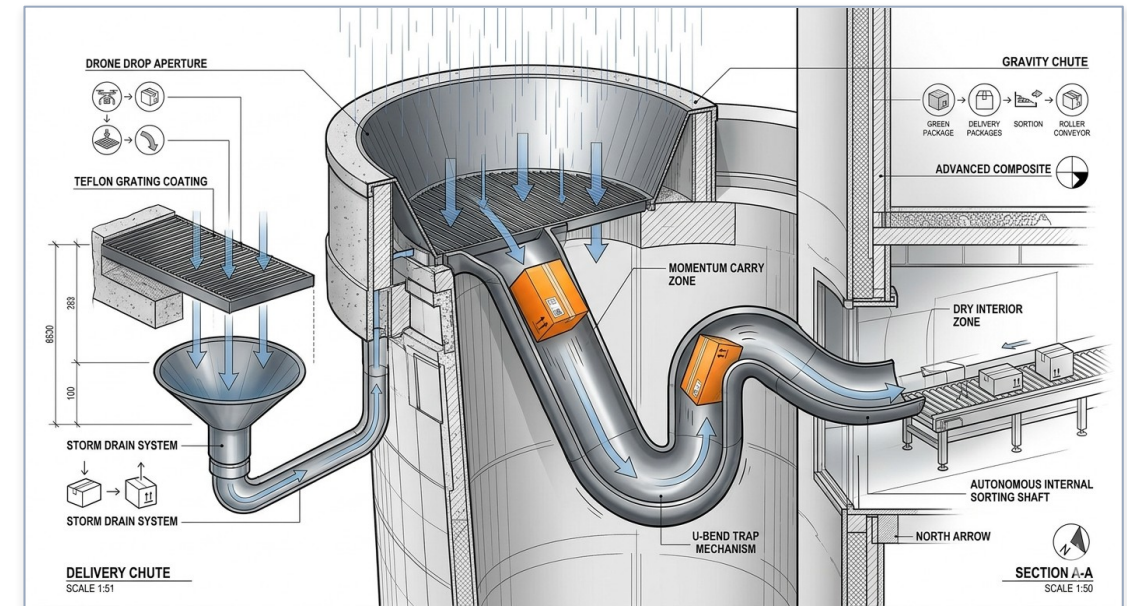
Design — Rooftop Flight Deck

- Stable delivery zone with perforated wind-break louvers to tame rooftop turbulence.
- Aerodynamic catchment funnel sized for off-center hover drops.
- Secure storage lockers for high-value or oversized parcels awaiting retrieval.
- Hover-winch protocol: drone holds altitude Z, lowers parcel through funnel mouth, releases tether — never lands.
- Flight-path corridor geofenced into building airspace; multiple drones serialized on approach.



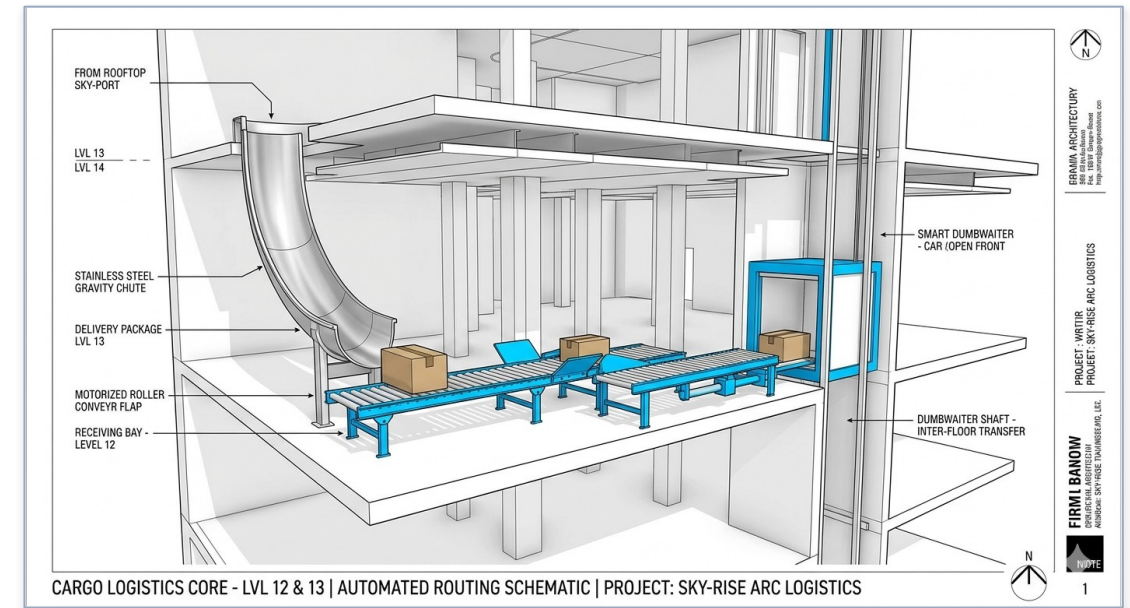
Design — Drop Funnel & S-Trap Drainage

- Drone drop aperture with Teflon-grated inner skin — parcel slides, water passes through.
- U-bend / S-Trap mechanism — passive drainage; rainwater diverts to the storm drain, parcel continues down the Momentum Carry Zone.
- Dry interior zone opens onto the autonomous internal sorting shaft and roller conveyor.
- Solves the rain-in-shaft constraint from content.md — no powered seals, no moving water doors.



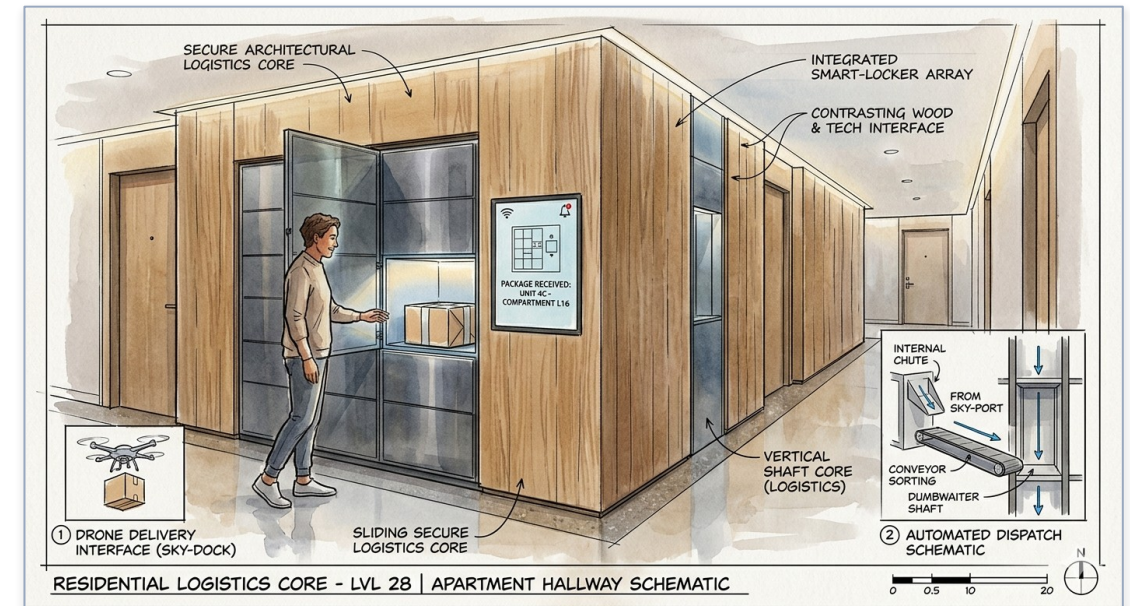
Design — Internal Sorting Core

- Stainless-steel gravity chute delivers parcels from the rooftop core to floor-level receiving bays.
- Motorized roller conveyor + diverter flaps route each parcel to its destination column.
- Smart dumbwaiter (open-front car in a dedicated shaft) lifts/lowers to the resident's floor.
- Embodies the passive-logistics principle from content.md: gravity + rollers + mechanical diverters in place of robotic arms.



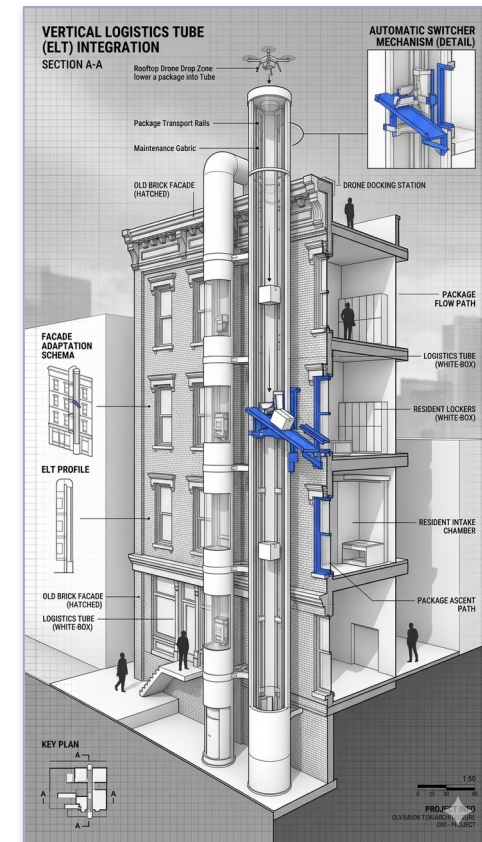
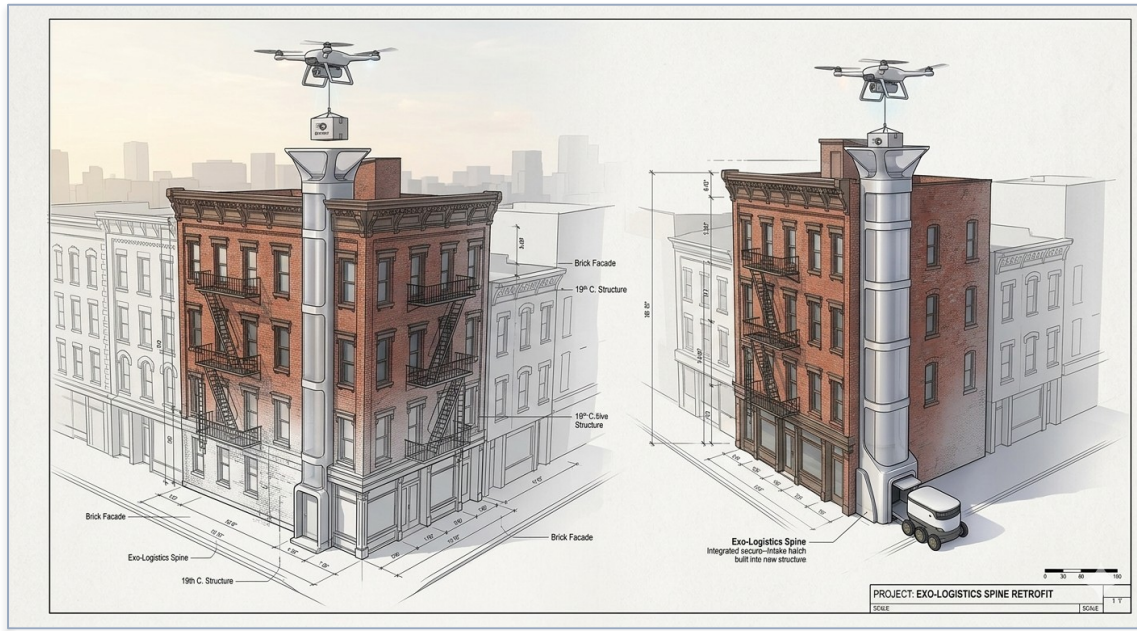
Design — Resident Smart-Locker Array

- Integrated smart-locker array built into the apartment-floor hallway wall — wood facade, glass + tech inserts.
- Sliding secure logistics core behind the lockers connects directly to the vertical shaft.
- Resident gets a push notification ("Package received: Unit 4C, Compartment L16"), authenticates at the screen, locker opens.
- Inset on the source drawing details the automated dispatch path: internal chute -> conveyor sorting -> dumbwaiter shaft.



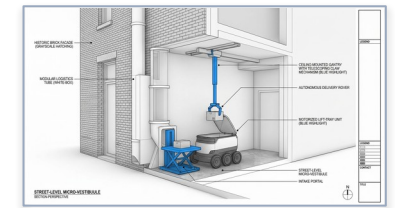
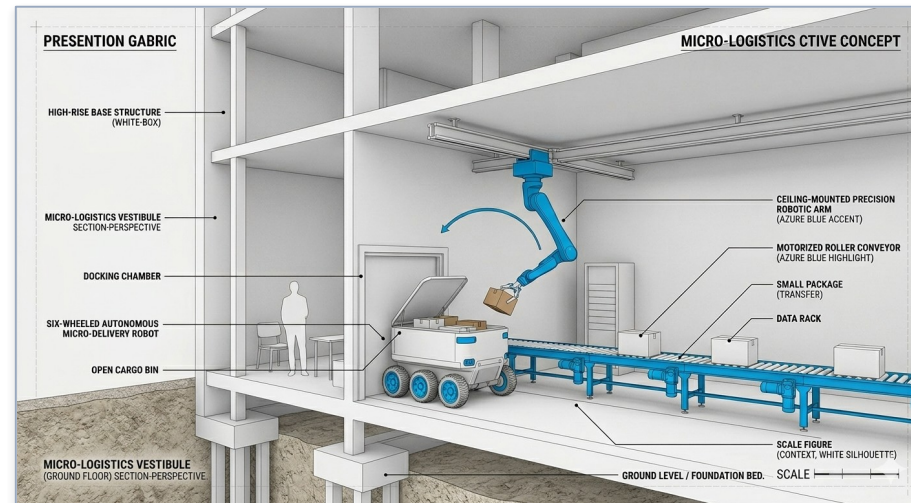
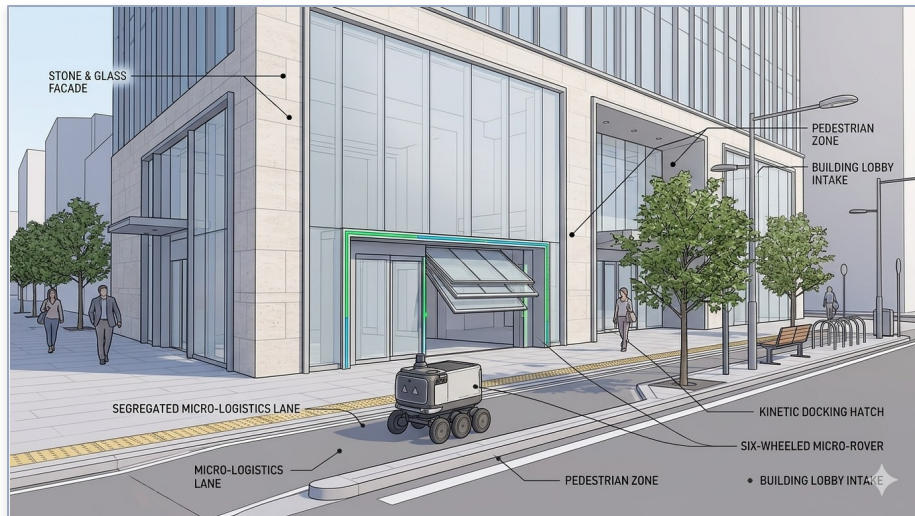
Design — Retrofit "Exo-Logistics Spine"

- Exterior modular tube clamped to the facade of historic buildings — no structural intervention to the protected envelope.
- Pneumatic decelerator inside the tube progressively brakes a free-falling parcel.
- Automatic switcher mechanism (right) diverts the parcel into each floor's resident intake chamber.
- Rooftop drone docking station caps the spine; a drone lowers the parcel into the tube mouth.



Design — Ground-Level Micro-Rover Interface

- Segregated micro-logistics lane along the sidewalk; rover approaches the kinetic docking hatch in the facade (left).
- Inside the vestibule (middle), a ceiling-mounted precision robotic arm lifts parcels off the rover's open cargo bin onto a motorized roller conveyor.
- Retrofit equivalent (right): street-level micro-vestibule with a telescoping claw + motorized lift-tray.



*Retrofit equivalent —
telescoping claw vestibule*

Implementation & Testing

- Subscale rooftop funnel + instrumented drop rig — characterize capture cone vs. wind speed.
- Single-floor mockup of internal sorting + smart-locker — measure throughput and jam rate.
- Exo-Spine pilot section (3 storeys) on a non-occupied historic structure — validate pneumatic deceleration and switcher routing.
- Rover <-> hatch integration with a partner micro-rover OEM.

KPI	Target
Drop capture success rate	<i>>= 99.5%</i>
Mean time, rooftop -> locker	<i><= 3 min</i>
Facade acoustic level	<i><= 55 dBA @ 10 m</i>
Rain ingress past S-Trap	<i>0 ml/h @ 25 mm/h rainfall</i>
Rover handoff cycle time	<i><= 45 s</i>

Part 3 — Roadmap / Project Plan

Roadmap / Project Plan

- *24-month phased plan — refine with real funding milestones.*

Phase	Months	Focus	Exit gate
P1 — R&D	1–4	Concept refinement, simulation, safety case	Design review #1
P2 — Subsystem prototypes	5–10	Funnel, S-Trap, switcher, hatch, rover dock	Bench-test sign-off
P3 — Integrated single-floor pilot	11–16	New-build mockup + Exo-Spine 3-storey rig	End-to-end KPI demo
P4 — Certification & approvals	17–20	FAA waiver, fire & building code, landmark commission	Permits granted
P5 — Commercial pilot	21–24	One new-build tower + one retrofit block	Paying first customer

Milestones, Deliverables & Work Packages

• Owners and exact months TBD.

WP	Title	Key deliverables	Lead	Month
WP1	Rooftop Hub	Flight Deck mock-up, hover-drop protocol spec	[name]	M8
WP2	Vertical Core	Sorting-shaft prototype, dumbwaiter integration	[name]	M12
WP3	Exo-Spine	Pneumatic-tube section, automatic switcher	[name]	M14
WP4	Rover Interface	Kinetic docking hatch + robotic-arm vestibule	[name]	M12
WP5	Software & Orchestration	Fleet manager, resident app, routing engine	[name]	M16
WP6	Regulatory & Safety	FAA waiver, fire/building/landmark approvals	[name]	M20
WP7	Pilot Deployment	One new-build + one retrofit operating	[name]	M24

Task Allocation & Timeline (24 months)

Work Package	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Owner					
WP1 Rooftop Hub																									[name]					
WP2 Vertical Core																										[name]				
WP3 Exo-Spine																									[name]					
WP4 Rover Interface																										[name]				
WP5 Software																									[name]					
WP6 Regulatory																										[name]				
WP7 Pilot Deployment																														[name]

Part 4 — Commercialization & Business Model

Market Analysis

Target market

- Luxury high-rise residential developers (new construction)
- Smart-city districts and master-planned communities
- Historic-district Business Improvement Districts commissioning the Exo-Spine retrofit
- Mixed-use towers with concierge service expectations

Customer profile

- Developer / building owner buys the infrastructure; residents are the end users
- Logistics providers (Amazon, FedEx, UPS, local couriers) are channel partners paying per-delivery

Bureaucracy & approvals

- FAA Part 107 + local airspace authority
- Building & fire code (UL listing for shafts)
- Landmark / historic-preservation commissions (retrofit only)

Market size

- *TAM: \$XX B global vertical-living logistics*
- *SAM: \$X B Tier-1 cities with drone-friendly regs*
- *SOM (Year 5): \$XXX M*

Marketing & growth strategy

- Lighthouse pilot with one flagship developer -> case study -> master-developer agreements
- "Logistics-ready building" certification co-marketed with developers
- API partnerships with major carriers for instant scale on Day 1 of a building going live

Competitor Analysis

Competitor	Their play	Where we win
Wing / Zipline / Matternet	Drone-to-doorstep delivery	We solve the last-meter (lobby -> door); they stop at the curb / yard
Starship / Serve / Coco	Sidewalk micro-rovers	We integrate with their rovers via a standard hatch — we are the building-side complement, not a competitor
Amazon Hub / InPost	Ground-floor parcel lockers	Our smart-lockers are on the resident's own floor, fed automatically — no lobby trip
Relay / Savioke	In-building delivery robots	We bypass elevators with a dedicated shaft; lower opex, higher throughput
Pneumatic-tube vendors	Hospital-grade tubes	We add aerial intake, weather handling, and resident-locker layer

- Our competitive advantage: only end-to-end, building-integrated stack from airspace to apartment door; weatherproof passive shafts; retrofit option (Exo-Spine) unlocks the entire historic-building market; modality-agnostic (drones, rovers, human couriers).

Budget, Pricing & Monetization

- *Cost structure — fill with real numbers.*

Line	Year 1	Year 3
R&D / engineering	\$X M	\$X M
Hardware (per-building install)	\$XXX k	\$XXX k
Software / cloud	\$X M	\$X M
Operations & support	\$X M	\$X M

- Pricing models: capex sale to developers / BIDs; SaaS orchestration per building per month; per-delivery fee to logistics partners; premium-resident subscription.
- *Unit economics: payback per building ~XX months; gross margin at scale ~XX%.*

Challenges & Business Risks

Risk	Likelihood	Impact	Mitigation
Urban drone-airspace regulation slips	High	High	Early FAA waiver process; partner with city pilot programs
Public acceptance (noise, privacy, drones overhead)	Med	High	Acoustic engineering at facade; opt-in resident comms; community design reviews
Insurance & liability for sky-dropped parcels	Med	Med	Capped payload, redundant tether, geofenced corridor, partner with logistics-insurance underwriter
Landmark-commission rejection of Exo-Spine	Med	High	Reversible / non-penetrating clamps; engage preservation architects from day 1
Weather window too narrow (wind, ice, lightning)	Med	Med	Hybrid sky + street modality — rovers cover the window where drones can't fly
Hardware reliability (jammed switcher, frozen rollers)	Med	Med	Passive-mechanical design philosophy; predictive maintenance via cloud telemetry
Capital intensity / long sales cycle to developers	High	Med	SaaS layer for recurring revenue; lighthouse-pilot case-study to compress sales cycle

Thank You

Questions & Discussion

A building-integrated logistics platform for new high-rises and historic retrofits.

[placeholder] contact: team@cargo-to-door.example | repo: github.com/<org>/cargo-to-door-design